

some users are still having trouble getting `--link-dest` to work properly. Make sure you use version 2.5.7 or later.

Update (2003.05.02): John Pelan writes in to suggest recycling the oldest snapshot instead of recursively removing and then re-creating it. This should make the process go faster, especially if your file tree is very large:

```
mv backup.3 backup.tmp
mv backup.2 backup.3
mv backup.1 backup.2
mv backup.0 backup.1
mv backup.tmp backup.0
cp -al backup.1/. backup.0
rsync -a --delete source_directory/ backup.0/
```

2003.06.02: OOPS! Rsync's `link-dest` option does not play well with J. Pelan's suggestion--the approach we previously had written above will result in unnecessarily large storage, because old files in `backup.0` will get replaced and not linked. Please only use Dr. Pelan's directory recycling if you use the separate `cp -al` step; if you plan to use `--link-dest`, start with `backup.0` empty and pristine.

10.9 Isolating the backup from the rest of the system

If you take the simple route and keep your backups in another directory on the same file system, then there's a very good chance that whatever damaged your data will also damage your backups. In this section, we identify a few simple ways to decrease your risk by keeping the backup data separate.

10.10 The easy (bad) way

In the previous section, we treated `/destination/` as if it were just another directory on the same filesystem. Let's call that the easy (bad) approach. It works, but it has several serious limitations:

- If your filesystem becomes corrupted, your backups will be corrupted too.
- If you suffer a hardware failure, such as a hard disk crash, it might be very difficult to reconstruct the backups.
- Since backups preserve permissions, your users--and any programs or viruses that they run--will be able to delete files from the backup. That is bad. Backups should be read-only.
- If you run out of free space, the backup process (which runs as root) might crash the system and make it difficult to recover.
- The easy (bad) approach offers no protection if the root account is compromised.